

HAT-P-36b

R-1866

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260604 · created 2026-08-03T10:12:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461195.7756 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.041
Transit depth (Rp/Rs)^2	2.45 +/- 1.29 [%]
Orbital Inclination (inc)	89.43 +/- 2.91
Airmass coefficient 1 (a1)	0.905 +/- 0.039
Airmass coefficient 2 (a2)	0.068 +/- 0.027
Scatter in the residuals of the lightcurve fit is	4.45 %
Best Comparison Star	#3 - [171, 461]
Optimal Aperture	3.49
Optimal Annulus	14.35
Transit Duration (day)	0.0986 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.041 vs 0.126 (deviation 0.76σ)
Duration fit vs published	0.0986 d vs 0.0925 d (deviation 1.27σ)
Mid-time O–C	-16.7 min
Depth SNR	3.8
Residual scatter	4.45 %

Light curve

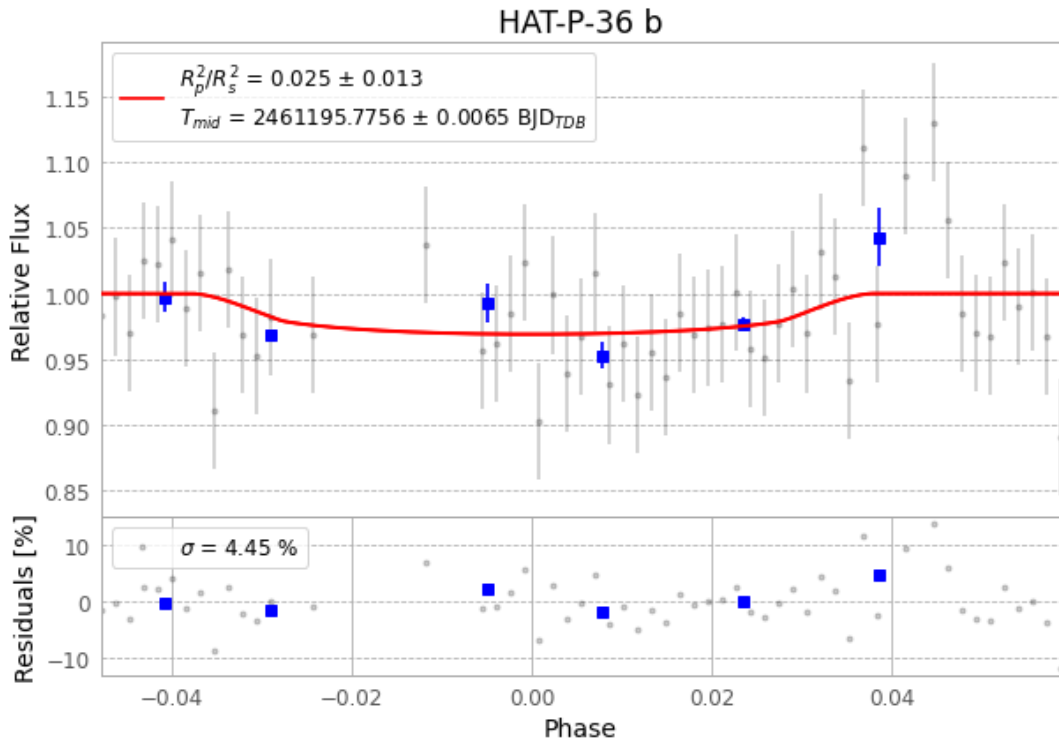


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-06-03.png (SHA-256 6ac0a6bc8688c345...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [238, 203], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9d473f7637a025d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), HATP-36260604050315.FITS → HATP-36260604082715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260604.log (db9054aed4ab...), FinalLightCurve_HAT-P-36 b_2026-06-03.png (6ac0a6bc8688...), FinalLightCurve_HAT-P-36 b_2026-06-03.csv (a8609b54780d...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 10:12Z.